

BONE FRACTURES

Recognition | Immobilisation | Splinting | Low-Resource Management

Clinical Guideline for AI Model Training Data — Version 1

Sources: WHO Emergency Care | MSF Clinical Guidelines | Hesperian Health Guides | ICRC War Surgery | WHO Trauma Care Manual

In conflict and low-resource settings, definitive surgical management may be unavailable for hours or days. The priority is correct recognition, prevention of further injury, pain reduction through immobilisation, and early identification of limb-threatening or life-threatening complications.

1. FRACTURE CLASSIFICATION

Type	Description	Risk Level
Closed (simple)	Bone broken; skin intact over fracture site	Lower — infection risk minimal
Open (compound)	Bone breaks through skin OR wound communicates with fracture	HIGH — infection, blood loss, bone loss
Greenstick	Incomplete fracture — common in children; bone bends rather than fully breaks	Lower — often stable
Comminuted	Bone shattered into multiple fragments	Higher — harder to stabilise
Stress / fatigue	Repeated loading causes hairline crack — no trauma	Lower acutely; requires rest
Pathological	Fracture through diseased bone (TB, tumour) with minimal or no trauma	Refer — underlying disease

2. GENERAL SIGNS OF FRACTURE

Sign	Description	Notes
Pain	Localised pain directly over bone, worsened by movement or palpation	Most reliable sign
Swelling	Rapid swelling around fracture site	More prominent within 30–60 minutes
Deformity	Visible angulation, shortening, or rotation of limb	Indicates displacement
Crepitus	Grating sensation or sound on gentle movement	Do NOT deliberately test — unnecessary pain
Loss of function	Unable to bear weight or use limb normally	Cannot exclude fracture if function retained
Neurovascular signs	Numbness, tingling, pale or blue fingers/toes, absent pulse distal to fracture	EMERGENCY — vascular injury

3. IMMOBILISATION PRINCIPLES (LOW-RESOURCE)

Immobilisation reduces pain, prevents further injury, and reduces blood loss from bone ends moving in tissue. Improvised splints from available materials are effective when applied correctly.

- Splint the fracture in the position found — do NOT attempt to realign unless neurovascular compromise.
- Immobilise the joint ABOVE and BELOW the fracture.
- Pad all bony prominences before applying splint to prevent pressure sores.
- Check circulation, sensation, and movement (CSM) before and after splinting.

- Never apply splint so tightly it cuts off circulation — fingers/toes should remain warm and pink.
- Elevate the limb if possible to reduce swelling.

3.1 Improvised Splint Materials

Material	How to Use	Notes
Straight sticks or branches	Place alongside limb, pad with cloth, secure with strips of clothing	Most available in field
Rolled newspaper or cardboard	Roll tightly or fold; mould to limb contour; wrap with cloth strips	Effective for arms, hands, feet
Folded blanket or clothing	Fold into thick pad; wrap around limb and secure	Soft splinting — good for children
Sam splint (aluminium + foam)	Mould to limb; excellent versatility	Include in first aid kits if possible
Traction splint (femur fractures)	Can be improvised with two long sticks and bandage loops	Required for femur fractures

4. MANAGEMENT BY FRACTURE LOCATION

4.1 Upper Limb Fractures

Location	Immobilisation Method	Key Risks
Clavicle (collar bone)	Broad arm sling supporting the arm. Figure-of-eight bandage if available.	Usually heals without surgery
Humerus (upper arm)	U-slab splint with sling or collar-and-cuff. Hang arm in sling.	Radial nerve injury — check wrist extension
Forearm (radius/ulna)	Splint from below elbow to palm; arm in sling at 90 degrees.	Check median and ulnar nerve function
Wrist	Splint palm to mid-forearm; elevate hand in sling.	Scaphoid fractures often subtle — treat any wrist injury as fracture
Hand / fingers	Buddy-strap finger to adjacent finger OR splint in position of function.	Preserve hand function position

4.2 Lower Limb Fractures

Location	Immobilisation Method	Key Risks
Pelvis	Do NOT move unless life-threatening. Wrap pelvis with sheet or belt to compress. Log-roll only.	LIFE-THREATENING haemorrhage — up to 3–4 L blood loss
Femur (thigh bone)	Traction splint essential — traction reduces blood loss and pain. Improvise with poles if no commercial splint.	1.5–2 L blood loss; shock common
Patella (kneecap)	Leg in extension; splint full length of leg.	Unable to extend knee actively
Tibia / fibula (lower leg)	Long leg splint from groin to foot; pad well.	Open fractures common — high infection risk
Ankle	Splint in neutral position (foot at 90 degrees). Pad well.	Do not force into position
Foot / toes	Soft padding; buddy-strap toes; do not bear weight.	Lisfranc injury often missed

4.3 Spine Fractures

- Suspect spinal fracture in: fall from height, road traffic accident, blast injury, diving injury, or any unconscious trauma patient.
- Immobilise the entire spine — do NOT flex, extend, or rotate the neck or back.
- Log-roll technique with minimum 3 helpers to move patient: keep spine in straight alignment throughout.
- Improvised cervical collar: rolled towels, newspaper, or clothing wrapped around neck and secured — must not restrict breathing or blood flow.

- Transport on rigid surface (door, stretcher, flat board) with spine maintained in neutral alignment.
- Do NOT remove a motorcycle or military helmet if worn — leave in place unless airway threatened.

DO NOT attempt to straighten or 'pop back' a deformed bone — can sever blood vessels and nerves.

Open fractures: cover wound with clean wet cloth; do NOT push bone back in; refer urgently.

Femur or pelvis fracture — assume major blood loss; lie patient flat; treat for shock.

Pale, cold, pulseless limb distal to fracture = vascular injury — SURGICAL EMERGENCY.

5. OPEN (COMPOUND) FRACTURES — WOUND CARE

STEP 1	CONTROL BLEEDING
	Apply gentle pressure around (NOT over) the exposed bone. Use clean cloth. Do NOT remove foreign material from deep in wound.
STEP 2	COVER WOUND
	Place a clean, moist dressing over the wound. Saline-soaked gauze preferred. Prevent drying of exposed tissue.
STEP 3	IMMOBILISE
	Splint as for closed fracture. Do NOT attempt to reduce the fracture.
STEP 4	DO NOT IRRIGATE FORCEFULLY
	Gentle irrigation with clean water only if gross contamination. No scrubbing.
STEP 5	REFER URGENTLY
	All open fractures require surgical debridement and wound care within 6 hours if possible to prevent osteomyelitis.

6. COMPARTMENT SYNDROME — EMERGENCY RECOGNITION

Compartment syndrome occurs when pressure within a muscle compartment rises to a level that cuts off blood supply. It can follow fractures, crush injuries, or tight bandaging. This is a surgical emergency.

Sign	Description
Pain	Severe pain disproportionate to injury; pain on passive stretching of muscles in affected compartment
Pressure / tension	Tense, woody-hard feeling to touch over muscle compartment
Pallor	Pale skin distal to injury
Pulselessness	Absent or diminished pulse distal to injury (late sign)
Paraesthesia	Numbness or tingling distal to injury (early nerve ischaemia)
Paralysis	Loss of movement distal to injury (late — severe nerve/muscle damage)

Compartment syndrome requires immediate surgical fasciotomy — cannot be treated in the field.

NEVER apply circular bandaging, plaster, or compression dressings to a swollen limb if compartment syndrome is suspected.

Loosen any bandages or splints immediately if compartment syndrome signs appear.

7. RED FLAGS — CRITERIA FOR URGENT REFERRAL

Red Flag Criterion	Clinical Significance / Action
Open fracture	All open fractures — surgical debridement required within 6 hours
Femur / pelvis fracture	Major haemorrhage risk — shock likely; urgent surgical stabilisation
Spine fracture	Any spinal injury — risk of permanent paralysis with further movement
Vascular compromise	Pale, cold, pulseless, or blue limb — vascular surgical emergency
Compartment syndrome	Woody-hard compartment + disproportionate pain — fasciotomy required
Neurovascular deficit	Weakness, numbness or absent pulse below any fracture
Fat embolism	Sudden breathlessness, confusion, or petechiae after long bone fracture

REF. REFERENCES

[1] MSF Clinical Guidelines — Trauma and Surgery. MSF, 2022. <https://medicalguidelines.msf.org>

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[4] Hesperian Health Guides. Where There Is No Doctor. Berkeley: Hesperian Foundation.

[5] WHO. Guidelines for Essential Trauma Care. Geneva: WHO, 2004.

[6] American Academy of Orthopaedic Surgeons. Emergency Care and Transportation of the Sick and Injured. 10th ed.

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